

Team: USF SUAS (University of South Florida)

1. ODLIC Pipeline

The team uses a Raspberry Pi computer with Python software for their imaging system. They chose the Raspberry Pi Camera Module 3 because it has autofocus. The camera is placed on a servo (a small motor) that can move and point the camera at things on the ground. For localization, the video did not show a clear formula. But based on what they showed, they probably use the drone's GPS position, its height, and the camera's angle together to guess where the object is on the ground. They did not explain the exact method or name a specific algorithm.

2. Communication Systems

The team uses two different links between the drone and the ground station. The first one is DragonLink with ArduPilot and Mission Planner. This link sends commands to the drone and receives telemetry (flight data) back. They picked it because it is strong/reliable and has a lot of documentation online. The second link is Ubiquiti AirMax, used to send bigger data like images. They said they had some problems with this link and were maybe thinking about changing it. The onboard computer is a Raspberry Pi, and it connects to the ground station using the AirMax link.

3. Additional Insights

The autopilot system is ArduPilot, running on a Pixhawk (Racer) controller. It handles stability, mission planning, GPS navigation, autonomous flight, and dropping the payload. The plane body is an Apprentice STS airframe, chosen because parts are easy to find, it is light, it has good lift, and it can fly for a long time. For dropping the payload, they made a simple part using a servo motor and a rolled-up string. This part can be placed anywhere on the plane, and they could even add more than one if needed. For avoiding obstacles, they use two RTL-SDR devices, probably to detect signals from nearby planes. In testing, they flew 44 times total with a 100% success rate and no crashes or damage.